



Systemes Numériques / FPGA

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Sommaire



Lesson 1. Introduction aux FPGA et logique combo



Lesson 2. Logique sequentielle



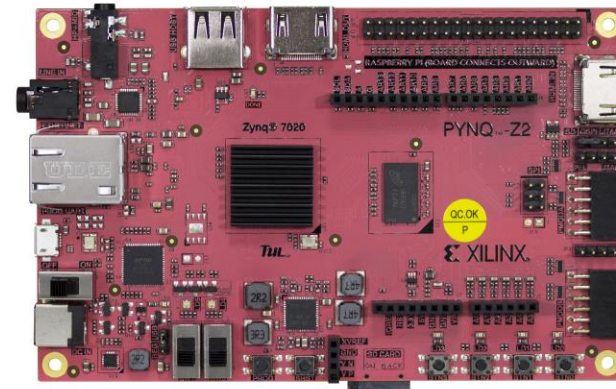
Lesson 3. Interfaces



Lesson 4. Pipelines



Lesson 5. Fun stuff



Introduction

FPGA interfaces:

- Datapath
- Pipelining



Datapath

- Part where the FPGA does calculations
- Represent data as : Integers, Decimal Fixed point, Binary Fixed Point, Floating Point, Logarithmic representation

Datapath

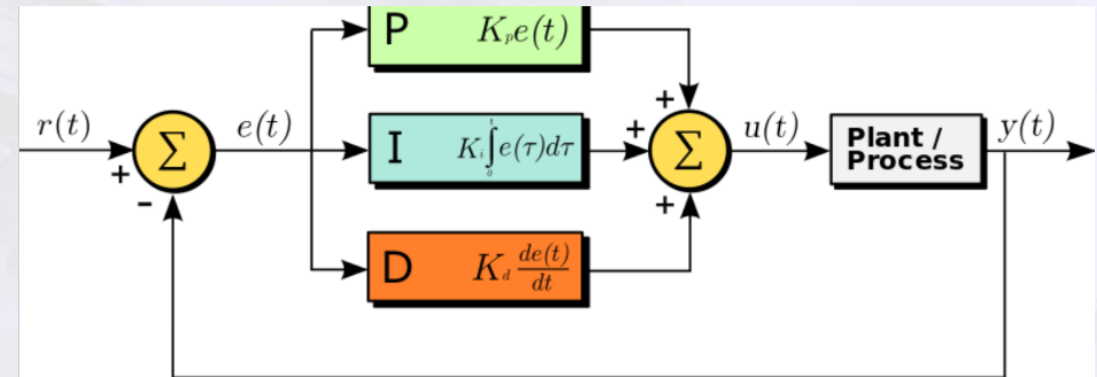
- Two metrics: dynamic range and precision
- Integers : Signed or unsigned
- How to represent -0.25 then ?
- Fixed point : signed or unsigned, defined as $Q_x.y$ or $S_x.y$. x is number of bits of integer part, y is number of bits in the fractional part.
- When everything else fails
- Floating point : signed, super large dynamic range at the price of a variable accuracy.
- Logarithmic: poor guy's FP, eases multiplication/division operations.

Datapath – Fixed point

- Use Q/S notation (TI style)
- Addition Align the integral parts of the numbers before adding/subtracting
- Multiplication is more complicated.
- Eg $S1.14 * S2.2 = S3.16$ number. $(-2 \text{ to } 2-2^{-14}) * (-4 \text{ to } 4-2^{-2}) \Rightarrow -8 \text{ to } 8$ range worst case. May need to saturate to avoid overflow or add 1 bit to the integer part.
- Truncation of the fractional part is allowed at the price of precision loss. *Integral part must not be truncated.*

Datapath Design

- Design a PID regulator
- Input of ADC is 16 bits signed, Output is 16 bits
- Architecture drawing
- Pipelining



Datapath Design - Approximation

- How to compute $|z| = \sqrt{x^2 + y^2}$?
- Naïve method with 16 bits -> Square root and squares
- Alpha-max+Beta-min algorithm -> Approximate

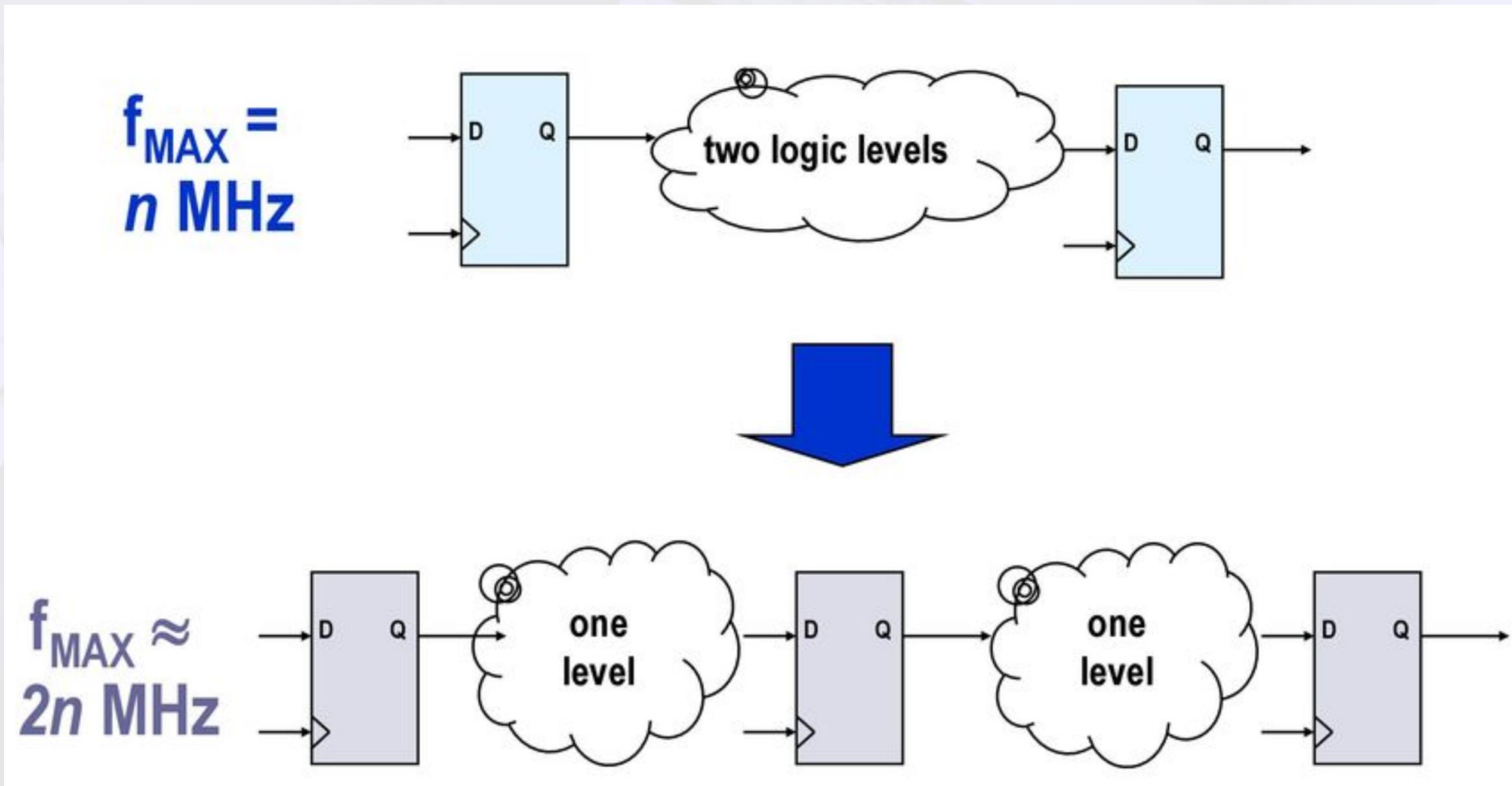
α	β	Largest error (%)	Mean error (%)
1/1	1/2	11.80	8.68
1/1	1/4	11.61	3.20
1/1	3/8	6.80	4.25
7/8	7/16	12.50	4.91
15/16	15/32	6.25	3.08
α_0	β_0	3.96	2.41

- Source: https://en.wikipedia.org/wiki/Alpha_max_plus_beta_min_algorithm

Datapath Design – Mix with control

- How to do a FIR filter ?
- Assume 16 bits signed inputs
- Output value length ?
- What is the latency of the a 1024 taps FIR ?

Datapath Design – Pipelining

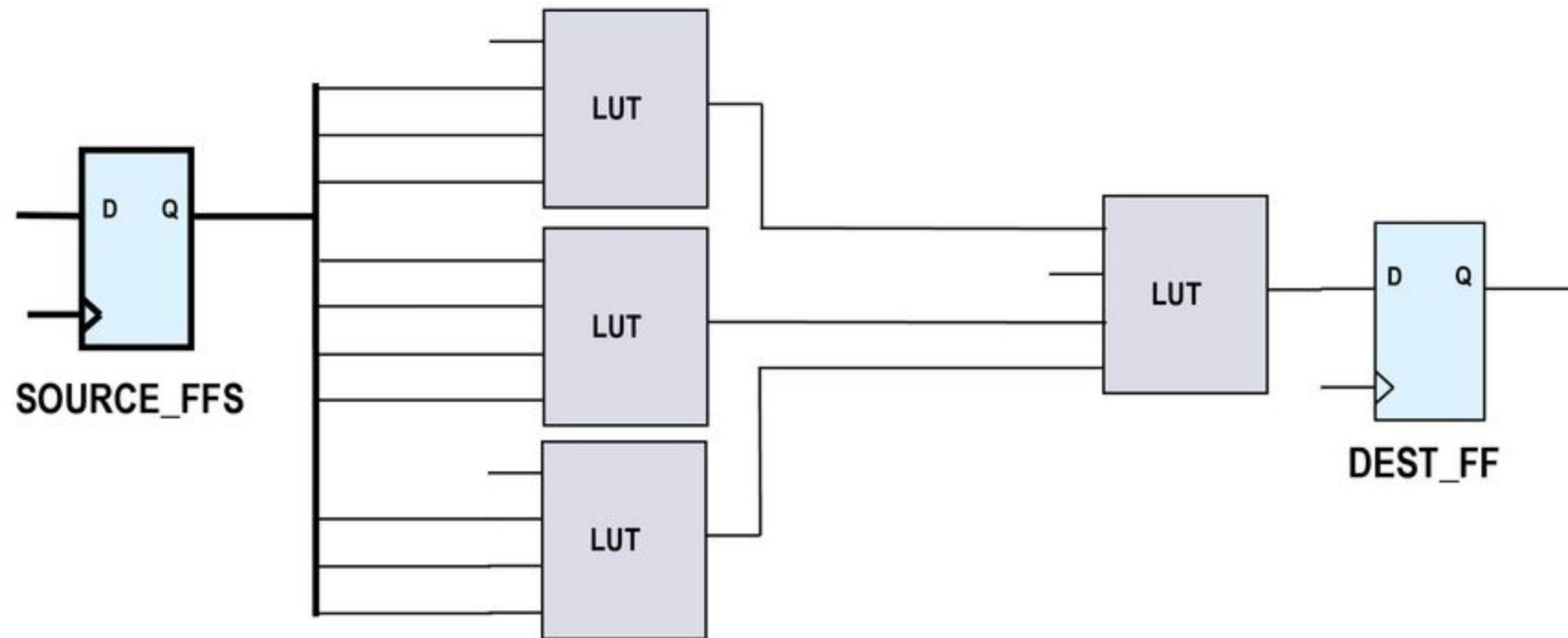


Reference : <https://slideplayer.com/slide/13517185/>

Datapath Design – Pipelining

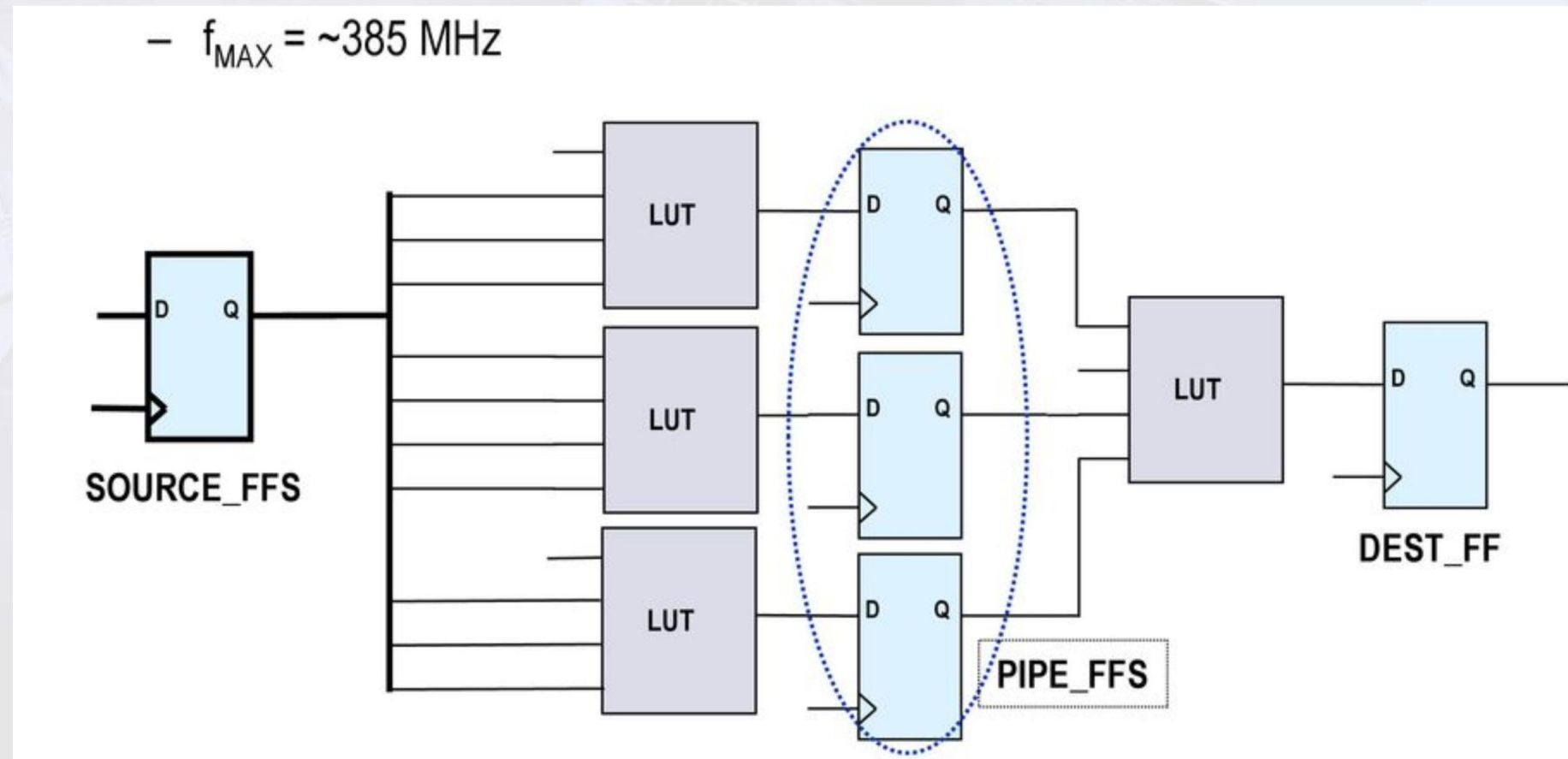
From this:

– $f_{\text{MAX}} = \sim 233 \text{ MHz}$



Datapath Design – Pipelining

To this:



Datapath Design – Pipelining

What's wrong with the pipelining here ?

